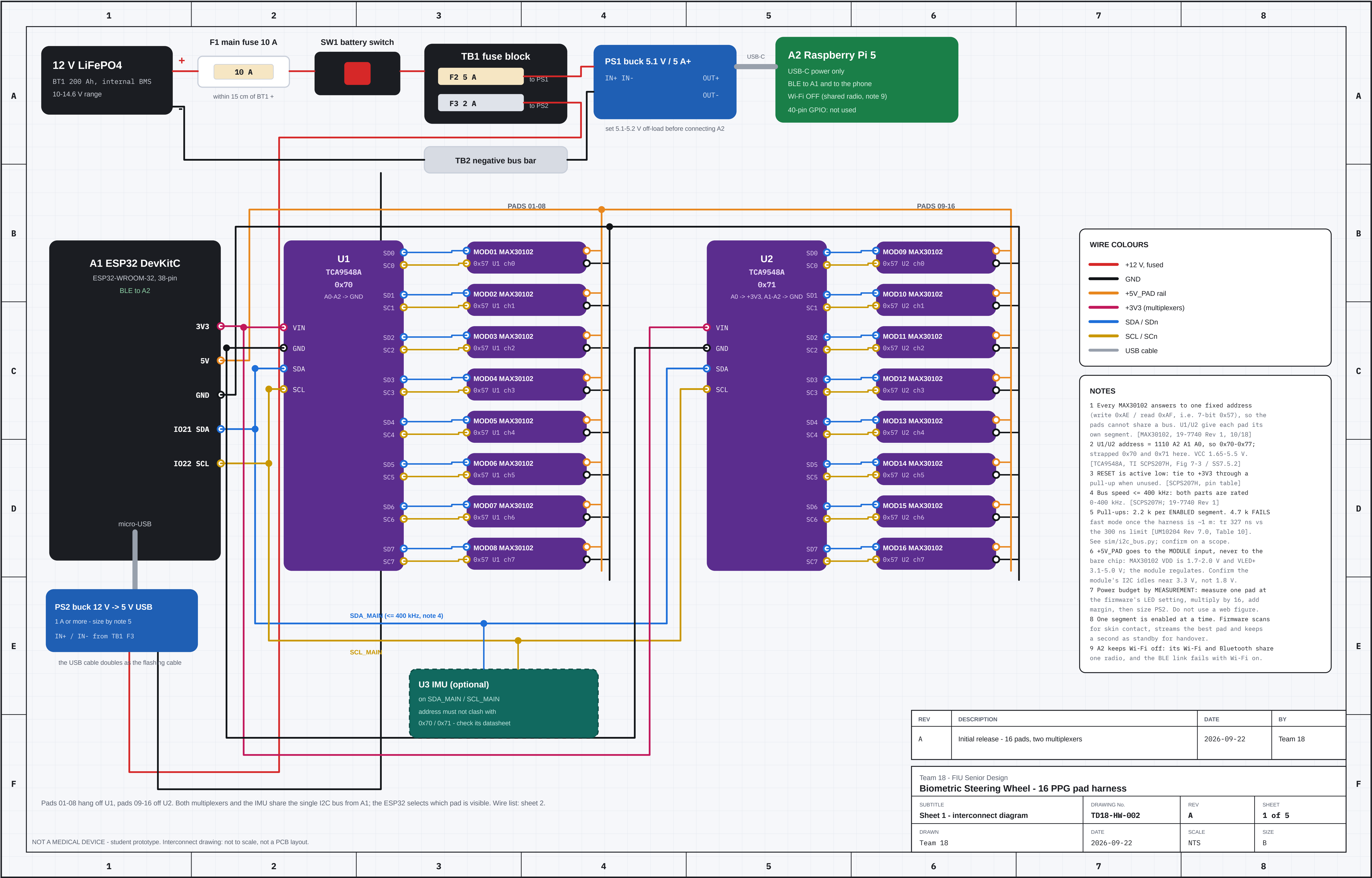
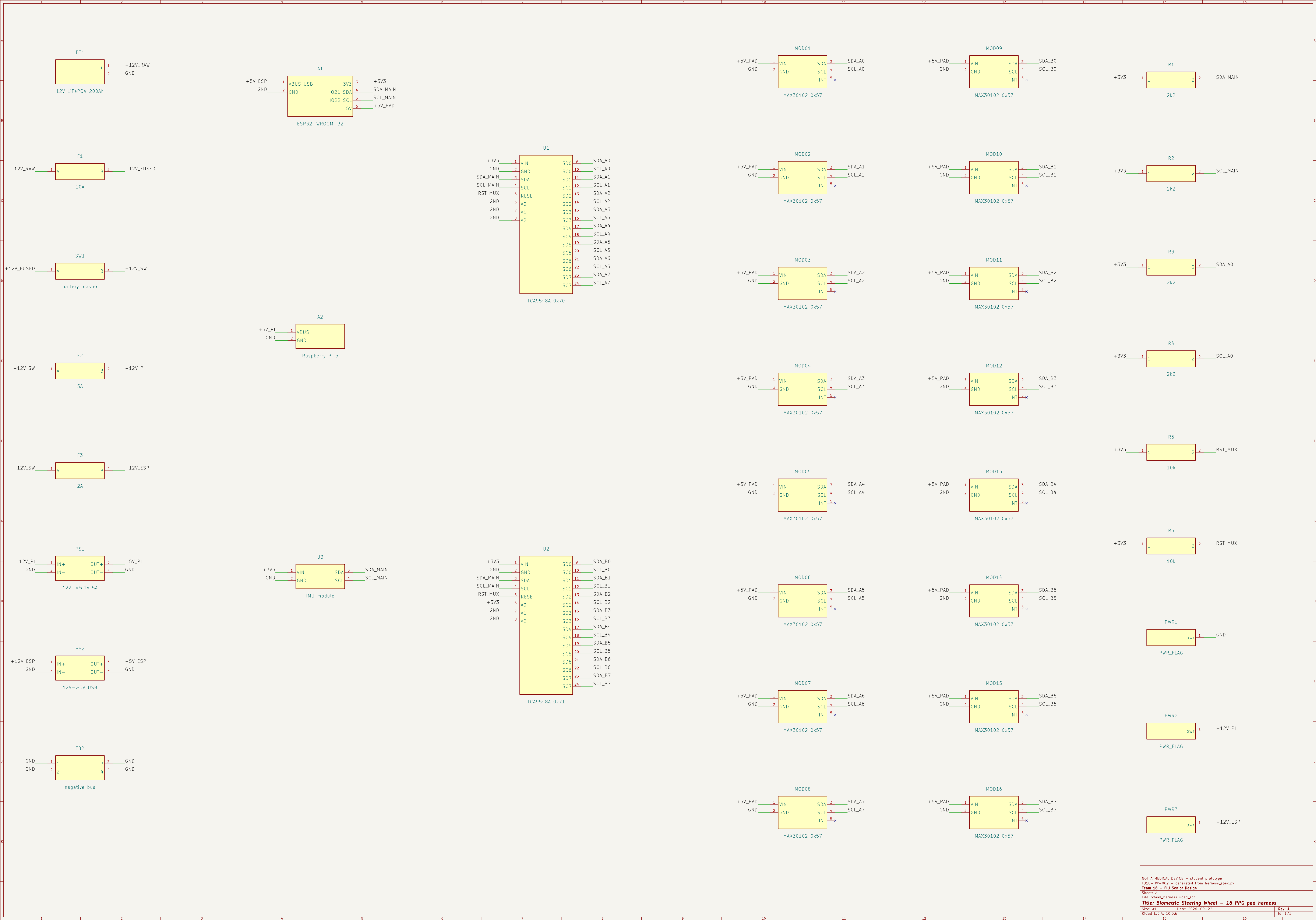




1		2		3		4		5		6		7		8				
A	PARTS LIST															A		
	REF	QTY	DESCRIPTION	KEY RATINGS / NOTES														
	BT1	1	12 V LiFePO4 battery, 200 Ah, internal BMS	10-14.6 V working range														
	F1	1	Fuse + holder, 10 A	within 15 cm of BT1 +														
	SW1	1	Battery master switch	disconnects the whole system														
	TB1	1	Fused distribution block, 2 ways	F2 5 A -> PS1, F3 2 A -> PS2														
	TB2	1	Negative bus bar	single return point														
	PS1	1	Buck converter 12 V -> 5.1 V, 5 A or more	set off-load before connecting A2														
	PS2	1	Buck converter 12 V -> 5 V USB	1 A or more; size by note 5, sheet 1														
	A1	1	ESP32-WROOM-32 DevKitC, 38-pin	I2C on IO21/IO22; BLE to A2														
B	A2	1	Raspberry Pi 5	USB-C power only; Wi-Fi off												B		
	U1, U2	2	TCA9548A 8-channel I2C switch module	0x70 / 0x71; VCC 1.65-5.5 V; 400 kHz														
	MOD01-16	16	MAX30102 pulse-oximetry module	0x57; module VIN, onboard regulator														
	U3	1	IMU module (optional)	on the main bus; address must not clash														
	R1-R4	a/r	Pull-up resistors 2.2 k	one pair per enabled segment (note 5)														
	R5, R6	2	Pull-up resistors 10 k	RESET of U1/U2 to +3V3 (note 3)														
	-	a/r	Harness wire, sleeving, strain relief	size from the measured current (note 7)														
	C	WIRE LIST															C	
		FROM	TO	NET	WIRE													
BT1 +		F1 in	+12V_RAW	red														
F1 out		SW1 1	+12V_FUSED	red														
SW1 2		TB1 in	+12V_SW	red														
TB1 F2 out		PS1 IN+	+12V_PI	red														
TB1 F3 out		PS2 IN+	+12V_ESP	red														
BT1 -		TB2	GND	black														
TB2		PS1 IN- , PS2 IN-	GND	black														
PS1 OUT+ / OUT-		A2 USB-C	+5V_PI	USB cable														
D	PS2 OUT	A1 micro-USB	+5V_ESP	USB cable												D		
	A1 3V3	U1 VIN, U2 VIN	+3V3	pink														
	A1 5V	MOD01-16 VIN	+5V_PAD	orange														
	A1 GND	U1/U2 GND, MOD01-16 GND, U3 GND	GND	black														
	A1 IO21	U1 SDA, U2 SDA, U3 SDA	SDA_MAIN	blue														
	A1 IO22	U1 SCL, U2 SCL, U3 SCL	SCL_MAIN	yellow														
	U1 SD0..SD7	MOD01..MOD08 SDA	SDA_A0..A7	blue														
	U1 SC0..SC7	MOD01..MOD08 SCL	SCL_A0..A7	yellow														
	U2 SD0..SD7	MOD09..MOD16 SDA	SDA_B0..B7	blue														
	U2 SC0..SC7	MOD09..MOD16 SCL	SCL_B0..B7	yellow														
E	U1 A0, A1, A2	GND	ADDR_U1 = 0x70	-												E		
	U2 A0	+3V3 (A1, A2 to GND)	ADDR_U2 = 0x71	-														
	U1 RESET, U2 RESET	+3V3 through 10 k	RST_MUX	-														
	F																F	
NOT A MEDICAL DEVICE - student prototype. Interconnect drawing: not to scale, not a PCB layout.																		
1		2		3		4		5		6		7		8				



	1	2	3	4	5	6	7	8																																																																								
A	WHERE AND WHEN THESE RESULTS CAME FROM <table><tr><th>ITEM</th><th>VALUE</th></tr><tr><td>Run finished</td><td>2026-09-22T17:59:34-04:00</td></tr><tr><td>Machine</td><td>hackerpc - Linux 7.2.5-3-omarchy</td></tr><tr><td>Schematic tool</td><td>10.0.6</td></tr><tr><td>Simulator</td><td>ngspice-47 : Circuit level simulation program</td></tr><tr><td>Renderer</td><td>rsvg-convert version 2.62.3</td></tr><tr><td>Python</td><td>3.14.7</td></tr><tr><td>Source</td><td>git luis/full-system @ f854593 - docs/hardware/</td></tr><tr><td>Drawing</td><td>TD18-HW-002 rev A - sheets 1-2 drawn, sheet 3 from KiCad</td></tr></table>				ITEM	VALUE	Run finished	2026-09-22T17:59:34-04:00	Machine	hackerpc - Linux 7.2.5-3-omarchy	Schematic tool	10.0.6	Simulator	ngspice-47 : Circuit level simulation program	Renderer	rsvg-convert version 2.62.3	Python	3.14.7	Source	git luis/full-system @ f854593 - docs/hardware/	Drawing	TD18-HW-002 rev A - sheets 1-2 drawn, sheet 3 from KiCad	DEVICE DATA: DOCUMENT, WHERE IT CAME FROM, WHAT WAS TAKEN <table><tr><th>PART</th><th>DOCUMENT</th><th>VALUE USED AND WHERE IT APPEARS</th></tr><tr><td rowspan="4">TCA9548A</td><td>TI SCPS207H</td><td>8 channels; 1110 A2 A1 A0 -> 0x70-0x77</td></tr><tr><td>MAY 2012, rev SEP 2024</td><td>Figure 7-3 / section 7.5.2 -> U1 0x70, U2 0x71</td></tr><tr><td>ti.com/lit/ds/symlink/tca9548a.pdf</td><td>VCC 1.65-5.5 V; fSCL 0-400 kHz; 5 V tolerant</td></tr><tr><td>fetchted 2026-09-22</td><td>RESET active low -> R5/R6 10 k to +3V3</td></tr><tr><td rowspan="2">MAX30102</td><td>Maxim 19-7740 Rev 1</td><td>write 0xAE / read 0xAF = 7-bit 0x57 -> note 1</td></tr><tr><td>10/18</td><td>VDD 1.7-2.0 V, VLED+ 3.1-5.0 V -> note 6</td></tr><tr><td rowspan="3">I2C bus</td><td>mirror: download.mikroe.com</td><td>fSCL 0-400 kHz -> note 4</td></tr><tr><td>(analog.com blocked the automated download)</td><td>FIFO depth 32 samples -> read in bursts</td></tr><tr><td>NXP UM10204 Rev 7.0</td><td>fast mode tr <= 300 ns, Cb <= 400 pF</td></tr><tr><td rowspan="2">IMU</td><td>1 October 2021, Table 10</td><td>-> the pull-up limit checked in SPICE</td></tr><tr><td>nxp.com/docs/en/user-guide/</td><td></td></tr><tr><td></td><td>not verified</td><td>address unknown until the module is chosen; it must not clash with 0x70 / 0x71</td></tr></table>				PART	DOCUMENT	VALUE USED AND WHERE IT APPEARS	TCA9548A	TI SCPS207H	8 channels; 1110 A2 A1 A0 -> 0x70-0x77	MAY 2012, rev SEP 2024	Figure 7-3 / section 7.5.2 -> U1 0x70, U2 0x71	ti.com/lit/ds/symlink/tca9548a.pdf	VCC 1.65-5.5 V; fSCL 0-400 kHz; 5 V tolerant	fetchted 2026-09-22	RESET active low -> R5/R6 10 k to +3V3	MAX30102	Maxim 19-7740 Rev 1	write 0xAE / read 0xAF = 7-bit 0x57 -> note 1	10/18	VDD 1.7-2.0 V, VLED+ 3.1-5.0 V -> note 6	I2C bus	mirror: download.mikroe.com	fSCL 0-400 kHz -> note 4	(analog.com blocked the automated download)	FIFO depth 32 samples -> read in bursts	NXP UM10204 Rev 7.0	fast mode tr <= 300 ns, Cb <= 400 pF	IMU	1 October 2021, Table 10	-> the pull-up limit checked in SPICE	nxp.com/docs/en/user-guide/			not verified	address unknown until the module is chosen; it must not clash with 0x70 / 0x71	A																					
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F					<table><tr><th>REV</th><th>DESCRIPTION</th><th>DATE</th><th>BY</th></tr><tr><td>A</td><td>Initial release - 16 pads, two multiplexers</td><td>2026-09-22</td><td>Team 18</td></tr></table> <table><tr><td colspan="4">Team 18 - FIU Senior Design</td></tr><tr><td colspan="4">Biometric Steering Wheel - 16 PPG pad harness</td></tr><tr><td>SUBTITLE</td><td>DRAWING No.</td><td>REV</td><td>SHEET</td></tr><tr><td>Sheet 4 - verification record</td><td>TD18-HW-002</td><td>A</td><td>4 of 5</td></tr><tr><td>DRAWN</td><td>DATE</td><td>SCALE</td><td>SIZE</td></tr><tr><td>Team 18</td><td>2026-09-22</td><td>NTS</td><td>B</td></tr></table>				REV	DESCRIPTION	DATE	BY	A	Initial release - 16 pads, two multiplexers	2026-09-22	Team 18	Team 18 - FIU Senior Design				Biometric Steering Wheel - 16 PPG pad harness				SUBTITLE	DRAWING No.	REV	SHEET	Sheet 4 - verification record	TD18-HW-002	A	4 of 5	DRAWN	DATE	SCALE	SIZE	Team 18	2026-09-22	NTS	B	F																																							
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NOT A MEDICAL DEVICE - student prototype. Interconnect drawing: not to scale, not a PCB layout.

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A	WHAT THE CHECKS FOUND, AND WHAT WAS CHANGED															A
	ID	DATE / FIXED IN	FOUND BY	WHAT WAS WRONG	WHY IT MATTERS	THE FIX	HOW IT WAS VERIFIED									
	F-01	2026-09-22 2ff2bc2	sim/i2c_bus.py, then ngspice	4.7 k pull-ups, as first drawn, give a 327 ns rise time on a ~1 m branch	Over the 300 ns fast-mode limit (UM10204 Table 10). The bus would still often work, but outside spec, and worse as the harness grows.	Sheets and parts list changed to 2.2 k; note 5 rewritten with the number and the citation.	ngspice: 152.8 ns at 2.2 k, 326.5 ns at 4.7 k on the same segment.									
	F-02	2026-09-22 93857ab	harness_spec.py self-check	The ESP32's USB input and its 5 V output pin were modelled as one pin	The spec refused to build: one pin cannot be on both +5V_ESP and +5V_PAD. On a real board these are different nets with a fuse/diode between them.	A1 now has VBUS_USB (power input, from PS2) and a separate 5V pin (power output, to the pads).	Spec self-check passes: 177 pins, no pin on two nets.									
	F-03	2026-09-22 d70a3fd	KiCad ERC	power_pin_not_driven on PS1 IN+ and PS2 IN+	A fuse is a passive part, so ERC could not see that the battery drives the rails past F2/F3. Real designs answer this with a power flag.	PWR2 and PWR3 flags added on +12V_Pi and +12V_ESP.	Those two errors gone from the ERC report.									
	F-04	2026-09-22 d70a3fd	KiCad ERC	pin_to_pin: three power outputs fighting on GND	The battery negative and both converter returns were all marked as drivers. Two drivers on one net is an error worth catching: on a real board it hides a wiring assumption.	Battery '-' and both OUT- pins are now passive; PWR1 alone drives GND.	No pin_to_pin violations remain.									
	F-05	2026-09-22 d70a3fd	KiCad ERC	pin_not_connected on pins that the drawing shows as wired	Stub wire ends were snapped to the 1.27 mm grid while the pins themselves sat at 135.84 mm: a 0.05 mm gap. Electrically open, visually identical. A picture can never catch this.	Symbol bodies are whole 2.54 mm steps and stubs now use the exact pin coordinate.	161 pin connections, all extracted by KiCad and matched against the spec.									
B																B
	F-06	2026-09-22 d70a3fd	KiCad ERC	lib_symbol_issues on all 38 symbols	The symbols were embedded in the sheet with no library to compare against, so KiCad could not confirm the placed symbol still matches its definition.	gen_schematic.py now also writes harness.kicad_sym and a sym-lib-table.	ERC: 0 errors, 0 warnings.									
C																C
	F-07	2026-09-22 684bf60	visual review of sheet 1	The F3 -> PS2 feed crossed over the PS1 block and its caption	Readability only, no electrical effect, but a wire that runs through a part is exactly what a reviewer marks up.	Rerouted: out of F3, down the gap before PS1, above the bus bar, then down a clear lane.	Re-rendered and checked; no wire now crosses a board on sheet 1.									
D	ASSUMPTIONS AND LIMITS - READ BEFORE TRUSTING A NUMBER															D
	Cable capacitance	60 pF/m ASSUMED	The rise-time result scales directly with it. Measure the finished harness with an LCR meter, or read tr on a scope, then re-run sim/i2c_bus.py --cpm.													
	Per-pad current	25 mA ASSUMED	Used only for the +5V_PAD ladder. Measure one module at the firmware's LED setting before sizing PS2.													
	Wire resistance	nominal AWG at 20 C	Ignores connectors, splices and temperature. Treat the simulated drop as a floor.													
	IMU address	unverified	ST's site blocked the automated download. Confirm against the datasheet of the module actually fitted; it must not clash with 0x70 / 0x71.													
	Module regulator	assumed present	5 V goes to module VIN, never to the bare chip (VDD 1.7-2.0 V). Confirm the module regulates and that its I2C idles near 3.3 V.													
	Schematic style	labels, not drawn wires	Connectivity is by net label, which is normal for a repetitive design and is what ERC and the netlist check verify.													
E	Scope															E
	electrical only	No PCB layout, no thermal or mechanical analysis, no EMC. Student prototype: NOT A MEDICAL DEVICE.														
F	WHAT THIS PACKAGE CONTAINS															F
	SHEET	CONTENT														
	1	Interconnect diagram: boards, wires, colour key, 9 notes														
	2	Parts list (15 lines) and wire list (21 nets), power-up checklist														
	3	KiCad schematic, generated from harness_spec.py, ERC clean														
	4	Verification record: environment, commands, results, datasheet evidence														
	5	This sheet: findings, fixes, assumptions and limits														
Sources	docs/hardware/ - drawing generator, spec, KiCad project, simulations															
Rebuild	python3 verify.py && python3 report_sheets.py && ./assemble.sh															

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